

SWARAJ KUMAR SAHOO

1st August, 2004

Jajpur Road, Jajpur, Odisha

📞 7205762913 ✉️ sswarajkumar848@gmail.com 🌐 swarajsahoo 📱 swarajsahoo

Education

National Institute of Technology, Rourkela

INTEGRATED MSc MATHEMATICS (CGPA: 8.06)

Nov 2022 – Present

Rourkela, Odisha

Relevant Coursework

- Data Structures & Algorithm
- Database Management
- Object Oriented Programming
- MYSQL
- Probability & Statistics

Projects

Audio Analyzer

Apr 2025

Material UI, APIs, Node.js, Express

GitHub

- * Built a real-time Audio Analyzer used by 100+ users for speech-to-text and sentiment analysis.
- * Implemented 9–12 backend routes and integrated 3 ML models, reducing audio classification time by 40%.
- * Designed an intuitive UI/UX and database pipeline to visualize call insights (speaker roles, hold time, dead air), improving analysis clarity and user adoption.

Credit Risk Analyzer

Mar 2025

Material UI, APIs, Node.js, Express

GitHub

- * Designed and implemented backend APIs using Node.js and Express to ingest, validate, and persist multi-factor loan applicant data into a structured database.
- * Engineered data pipelines to process and aggregate 50+ applicant records across credit history, income, and behavioral factors for reliable risk assessment.
- * Integrated ML-driven risk scores into the application workflow, enabling lenders to make faster, data-backed lending decisions.

Hazard Reporter

June 2025

React.js, Node.js, Express, MongoDB

GitHub

- * Designed a citizen safety platform enabling 300+ users to report hazards (potholes, accidents, etc.) directly to municipal bodies.
- * Integrated real-time notification system and progress tracking for 100+ reported hazards, reducing complaint resolution time by 35%.
- * Built 10+ backend routes and developed interactive dashboards for both users and authorities to monitor incidents.

Achievements & Certifications

- **April Cohort Hackathon (Apr 2025):** Winner among 50+ teams for an industry-focused product showcased to 5+ mentors.
- **SBI ideathon (Feb 2025):** Secured Top 1000 out of 50000+ teams SBI IDEATHON for project *Self-Improvement*.
- **Hack4Bengal 4.0 (2025):** Ranked in Top 100 of 1500+ teams for "Hazard Reporter," improving citizen reporting by 50%.

Technical Skills

- **Languages:** Python, C, C++, HTML/CSS, JavaScript
- **Operating Systems:** Windows, MacOS
- **Developer Tools:** VS Code, Sublime, AWS
- **Frameworks:** React.js, Node.js, Express.js, Mongoose, Tailwind, Bootstrap, Redux

Extracurricular Activities

AXIOM, Math Club – NIT Rourkela

May 2024 – May 2025

Secretary

- * Led a 30-member team and organized 5+ campus-wide math events with 200+ participants.

Algorithmic & Programming Society – NIT Rourkela

Aug 2024 – May 2025

Event Management Lead

- * Managed AlgoVision for 200+ participants from 4+ departments ensuring zero logistical delays.